How the Two Models Connect

The economic channels of the OG-Core × CLEWS soft link — and how each is represented in the model

numbers: the Philippine reference run · `ogclews-link` · July 9, 2026

The coupling problem

Two models, complementary gaps

CLEWS / OSeMOSYS

least-cost energy-system optimiser: technologies, dispatch, investment, and emissions

- **rich:** technology, season-by-season operation, emissions, land & water
- **gap:** energy demand is external — no price response, no households, no government budget, no distributional accounts

OG-Core

overlapping-generations general equilibrium: households, firms, and a government budget

- **rich:** behaviour, saving, fiscal policy, incidence, demography
- **gap:** no energy technology — firms do not use energy as a priced production input

The useful hand-off differs by direction:
quantities flow from the economy, **prices and physical outcomes** flow from the energy system.

The structural hand-off

OG-Core

the whole economy: households, firms, government

SOLVES

prices, wages & interest rates · investment, debt & taxes · who is richer or poorer

TAKES AS GIVEN

energy demand · the interest rate · carbon policy

(it doesn't model the energy system)

the economy → the energy system

WHAT'S NEEDED →

energy demand · interest rate

← **WHAT IT COSTS**

energy price · build cost · emissions

the energy system → the economy

CLEWS / OSeMOSYS

a least-cost model of the energy system (+ land & water)

SOLVES

what to build & run · emissions · the price of energy

TAKES AS GIVEN

energy demand · the interest rate · costs

(no households, jobs, or budget)

The coupled solution the energy the economy wants, at the system's price, matches what the system supplies

Each model is hollow exactly where the other is deep.

Quantities flow right (energy demand, the interest rate); prices and physical outcomes flow left (the energy price, build cost, emissions). The models stay separate and pass a few auditable signals.

Why a *staged soft* link, not a simultaneous hard link

A simultaneous hard link is structurally awkward:

- OG-Core moves through the transition period by period; the energy model optimises the full planning horizon at once
- neither model fits inside the other: OG-Core has no single objective the energy model could be nested in, and no shock can be injected mid-solve

So the models stay separate and the link runs in stages:

solve OG-Core baseline → solve CLEWS, extract signals → apply them to the economy; solve reform → send results back to CLEWS ⇢ iterate to a fixed point

Current implementation: the single pass (solid). The loop (dashed) is designed, and the return signals already exist, but the outer iteration is not yet implemented.

Signals, wedges, and the main price signal

The vocabulary: signal $\rightarrow$ wedge $\rightarrow$ channel

Signal an outcome read from a *solved* energy scenario: electricity price, grid investment, or emissions

Wedge the OG-Core parameter that carries it: τ^c, Z_m, α_I, $\rho(s,t)$, or $e(t,s,j)$

Channel one signal mapped to one wedge, with a diagnostic check for large or poorly defined mappings

The country-specific pieces, such as the electricity industry, are read from the country's own model at run time.

The main hand-off is the electricity price

The most important CLEWS signal is the **electricity price**: the levelized cost of a delivered unit — capital, running costs, and fuel, averaged over the solved system's generation. It is the right hand-off because it lets the economy do two things the energy model cannot do on its own:

1. It lets demand respond

In CLEWS, demand is an external input. In OG-CORE, households change energy use when prices change, subject to a subsistence floor.

2. It shows who pays

A single "system cost" is silent on who bears it. OG-CORE's households, by age and income, split the burden — and because energy is a partial necessity, a price rise falls hardest on the poor.

In later iterations, the models move toward agreement between the electricity price and the cost of supplying the demand generated by the economy.

The energy channel: two uses, one transmission

OG-Core firms use capital and labour, not energy. The electricity-price signal is therefore carried through the two places electricity appears in the accounts.

Firms

Electricity as an intermediate input, about three quarters of use in the Philippine accounts. The channel applies a cost-push to industry production costs, which sets the output sign.

Households

Electricity as final consumption. The channel applies a consumption-tax wedge on the energy good; when revenue recycling is on, the revenue is rebated as transfers.

The electricity industry's own electricity purchases are netted out to avoid double counting. The input side remains reduced-form until energy is added as a priced production input.

The eight channels

The eight channels at a glance

Channel	Dir.	Signal / lever	OG-Core object	First-order effect
Energy price	C→O	electricity price (levelized cost / index)	cost-push on firms + consumption wedge	output ↓; regressive incidence
Carbon price	policy	carbon price ($/t$CO_2$)	energy-wedge add-on + emissions penalty	demand ↓; emissions ↓
Public investment	C→O	grid & transmission build cost	public-investment share → public capital	public capital ↑; crowding-out, debt
Capital intensity	C→O	power-fleet capital-cost share	energy industry's capital share (γ_m)	energy price ↓; energy K ↓
Investment incentive	policy	tax-credit rate	cost of capital (investment tax credit)	capital reallocates into energy
Health	C→O	$PM_{2.5}$ emissions change	survival rates + worker productivity	mortality ≈ 0; labour productivity ↑
Discount rate	O→C	economy's return	planning discount rate	shifts the build mix via iteration
Energy demand	O→C	economy's activity	specified annual demand	changes capacity via iteration

four CLEWS→OG-Core · two policy levers applied once to both sides · two OG-Core→CLEWS

Energy price — one price, two economic effects

Firm cost-push

- electricity-intensive industries get a productivity haircut
- the haircut is scaled by each industry's electricity input share
- this is the output and wage side of the price shock

Household incidence

- direct household energy consumption gets a consumption wedge
- the wedge is diluted by electricity's share of the energy good
- revenue is rebated as transfers when revenue recycling is on

The two pieces are one composite channel in the coupled run. A separate cost-push number is a diagnostic decomposition; the mechanism itself applies both pieces before solving the economy.

Carbon price — one policy, counted once

carbon price
$50/t$CO_2$

CLEWS: emissions-penalty path
$\Rightarrow$ the build mix decarbonises

OG-CORE: $\Delta\tau^{c}_{i_e} =$ price $\times$ unit conversion
$\times$ carbon intensity $/\, p^{b}_{i_e}$

- one chosen price enters *both* models along the same policy path; it is not also inferred from the energy model's own implied carbon cost
- OG-CORE firms buy no energy, so only *household* energy carbon is priced on the macro side; industrial carbon is priced in CLEWS
- revenue can be recycled as transfers; applied by itself, this channel changes steady-state output by -0.021%

that level is illustrative because the conversion from CLEWS monetary units to the OG-CORE numéraire is not yet calibrated.

Public investment — grid costs enter through public capital

grid & transmission build cost (% of GDP) → public-investment share α_I $\rightarrow$ public capital K_g → K_g raises all industries' output crowding-out if debt-financed

- this is where crowding-out enters the linked model: debt-financed spending competes for the savings pool, raises r, displaces private investment
- public-grid capital and private-generation capital are different objects; using both is complementary, not double counting
- in the Philippine scenario the *extra* grid spending is ≈ 0 because the transition is mostly generation-side, so this channel has essentially no effect here; the steady-state output change rounds to $+0.000\%$. The mechanics scale with any non-zero spending

Capital intensity — why the sign is not automatic

Signal: CLEWS says the clean power fleet is more capital-heavy. In OG-CORE, this is represented as a higher energy-sector capital share, not as a new investment subsidy or spending program.

capital share rises factor mix changes → unit cost and electricity price fall → sector revenue can fall even if output rises

What to watch

Energy capital follows the sector's revenue and user cost. If the price drop is larger than the output gain, the sector can use less capital even though the technology is more capital-intensive.

Outputs to update

- energy capital share
- energy price
- energy output
- energy capital stock

This is a factor-mix and price channel. The capital-pull lever is the investment incentive, which lowers the cost of capital.

Investment incentive — the actual capital-pull lever

The lever that directly attracts capital is an **investment tax credit** for the energy industry.

$$\rho_m = \frac{r + \delta - \tau_m^b \delta_m^\tau - \tau_m^{\mathrm{inv}} \delta}{1 - \tau_m^b}$$

The credit lowers ρ_m, where γ_m is absent; capital demand shifts *out* at a given interest rate.

Capital intensity

changes the factor mix through γ_m; the solved effect on energy capital depends on energy price, output, and user cost.

Investment incentive

changes the cost of capital through ρ_m; capital reallocates *into* energy. This lever is defined but not in the Philippine reference run.

The lesson

Two levers, two different objects. Capital intensity changes the factor mix; the tax credit changes the cost of capital. They can move energy capital in opposite directions.

Health — mortality is small, productivity matters

cleaner power: $\text{PM}_{2.5}$ emissions -2.7% → only $\sim$8% reaches deaths ($M \approx 0.082$) → ≈ 95 fewer deaths a year

→ mortality: survival rates $\rho(s,t)$

→ morbidity: worker productivity $e(t,s,j)$

Scale adjustment

$43{,}951 \times 0.082 \times (-0.0265) \approx -95$ deaths. The adjustment keeps the emissions cut from being read as all pollution deaths.

Macro effect

Mortality changes are retiree-heavy, so steady-state output barely moves (-0.0003%). Working-age morbidity gives the near-term gain ($+0.19\%$).

Time profile: productivity lifts the transition; retiree-heavy survival effects dominate the long run.

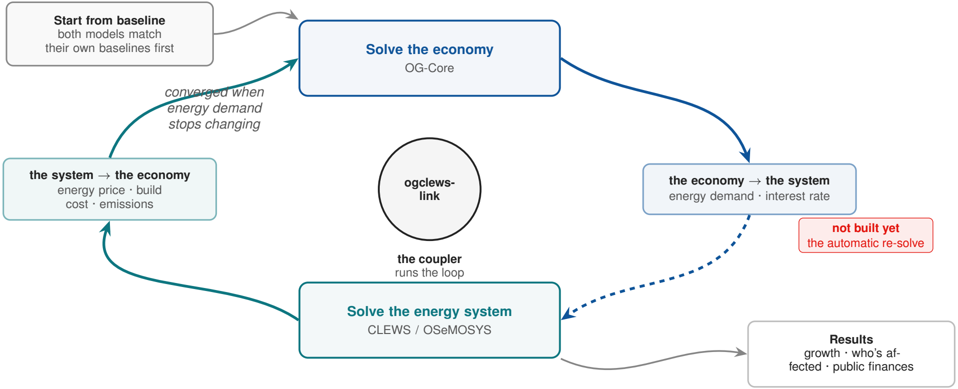
Return and demand go back to CLEWS

Two channels run *after* the economy solves and write ready-to-use CLEWS inputs:

- **discount rate**: the economy's equilibrium return r_p becomes the energy system's planning discount rate — so energy planning uses the *model-consistent* cost of capital. A lower return favours technologies with high up-front costs and long operating lives.
- **energy demand**: how activity changed (industry output or consumption, with an elasticity) scales the demand the system must meet. In a single pass the change is $\approx$ zero, so this channel mainly matters once the models iterate.

They affect the macro economy only once the loop iterates: the quantity link of the classic coupled energy–macro models (MESSAGE-MACRO, TIMES-MACRO), but feeding an economy with households and a government instead of a single representative agent.

Closing the loop



Iterate until energy demand stops changing. Current implementation: the single pass; the loop still needs damping.

The Philippine coupled run

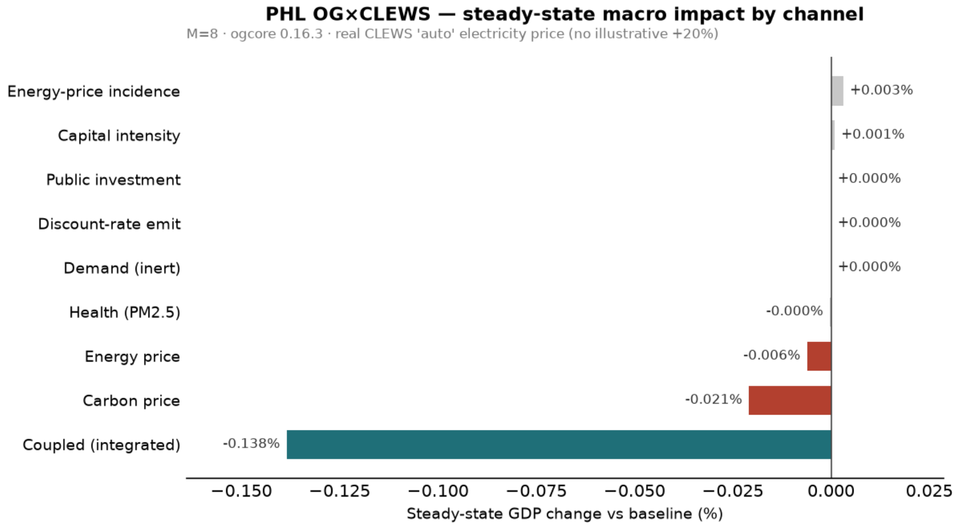
What is active in the coupled run

BEFORE THE ECONOMY SOLVES | **AFTER**

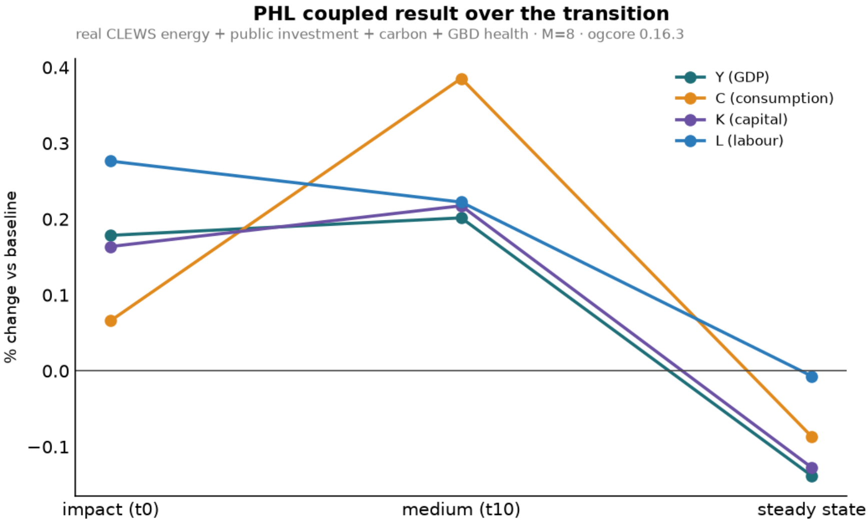
the scenario's real electricity price, sent two ways: industry cost-push + household wedge → public investment grid spending → public capital → carbon price \$50/tCO_2 (in the energy model) → health cleaner air → survival + productivity → **SOLVE** reform transition → send back: return + energy demand

- The real electricity price (its levelized cost) rises about 5% on average, so the energy-price channels stay small (-0.006% alone).
- Health carries the near-term gains; the steady-state contraction is larger than the sum of the standalone channels — the channels interact.
- The -0.021% carbon number is the household-wedge channel applied by itself, not an added second carbon price.

% change	steady	t_0	t_{10}
output Y	-0.138	$+0.179$	$+0.202$
consumption C	-0.087	$+0.066$	$+0.385$



Per-channel steady-state output impact, each run alone. Every standalone channel is small (carbon, -0.021%, is the largest contraction); the integrated coupled run (-0.138%) is well beyond their sum — the channels interact.



The coupled transition: $+0.179\%$ at t_0 and $+0.202\%$ by t_{10}, but -0.138% at the steady state. Working-age morbidity gains carry the near-term path; the long run combines the added retirees with the contractionary price channels.

Status

Implementation status

Implemented and checked

- eight one-signal $\rightarrow$ one-wedge channels
- single CLEWS→OG-Core pass on real price and health inputs
- separate model environments linked by auditable files
- numbers locked in the reference record and re-checked by tests

Remaining work

- **loop closure** — outer re-solve-and-iterate
- **monetary-unit conversion** — calibrate CLEWS money to the OG-Core numéraire
- **energy as a priced input** — structural firm-cost channel

Bottom line

A small, auditable library of mechanisms: each signal counted once, each effect attributable.